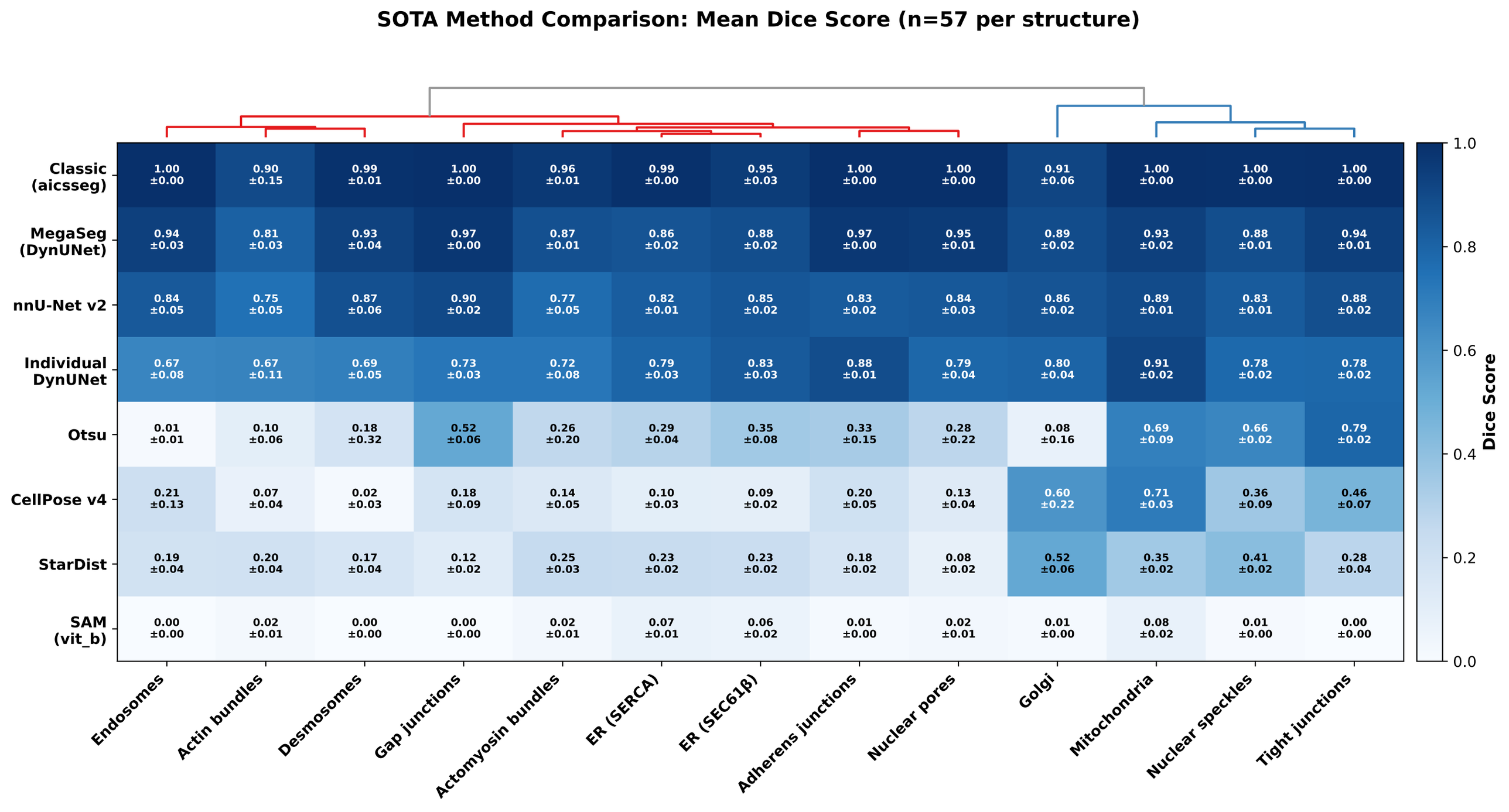

SOTA Method Comparison: Mean Dice Score (n=57 per structure)
Classic (aicsseg)
MegaSeg (DynUNet)
nnU-Net v2
Individual DynUNet
Otsu
CellPose v4
StarDist
SAM (vit_b)
1.00 ±0.00
0.90 ±0.15
0.99 ±0.01
1.00 ±0.00
0.96 ±0.01
0.99 ±0.00
0.95 ±0.03
1.00 ±0.00
1.00 ±0.00
0.91 ±0.06
1.00 ±0.00
1.00 ±0.00
1.00 ±0.00
0.94 ±0.03
0.81 ±0.03
0.93 ±0.04
0.97 ±0.00
0.87 ±0.01
0.86 ±0.02
0.88 ±0.02
0.97 ±0.00
0.95 ±0.01
0.89 ±0.02
0.93 ±0.02
0.88 ±0.01
0.94 ±0.01
0.84 ±0.05
0.75 ±0.05
0.87 ±0.06
0.90 ±0.02
0.77 ±0.05
0.82 ±0.01
0.85 ±0.02
0.83 ±0.02
0.84 ±0.03
0.86 ±0.02
0.89 ±0.01
0.83 ±0.01
0.88 ±0.02
0.67 ±0.08
0.67 ±0.11
0.69 ±0.05
0.73 ±0.03
0.72 ±0.08
0.79 ±0.03
0.83 ±0.03
0.88 ±0.01
0.79 ±0.04
0.80 ±0.04
0.91 ±0.02
0.78 ±0.02
0.78 ±0.02
0.01 ±0.01
0.10 ±0.06
0.18 ±0.32
0.52 ±0.06
0.26 ±0.20
0.29 ±0.04
0.35 ±0.08
0.33 ±0.15
0.28 ±0.22
0.08 ±0.16
0.69 ±0.09
0.66 ±0.02
0.79 ±0.02
0.21 ±0.13
0.07 ±0.04
0.02 ±0.03
0.18 ±0.09
0.14 ±0.05
0.10 ±0.03
0.09 ±0.02
0.20 ±0.05
0.13 ±0.04
0.60 ±0.22
0.71 ±0.03
0.36 ±0.09
0.46 ±0.07
0.19 ±0.04
0.20 ±0.04
0.17 ±0.04
0.12 ±0.02
0.25 ±0.03
0.23 ±0.02
0.23 ±0.02
0.18 ±0.02
0.08 ±0.02
0.52 ±0.06
0.35 ±0.02
0.41 ±0.02
0.28 ±0.04
0.00 ±0.00
0.02 ±0.01
0.00 ±0.00
0.00 ±0.00
0.02 ±0.01
0.07 ±0.01
0.06 ±0.02
0.01 ±0.00
0.02 ±0.01
0.01 ±0.00
0.08 ±0.02
0.01 ±0.00
0.00 ±0.00
Endosomes
Actin bundles
Desmosomes
Gap junctions
Actomyosin bundles
ER (SERCA)
ER (SEC61β)
Adherens junctions
Nuclear pores
Golgi
Mitochondria
Nuclear speckles
Tight junctions
Dice Score
1.0
0.8
0.6
0.4
0.2
0.0